

STATUTORY ENGINEERING NOTICE: BlueprintIQ provides preliminary automated computer-vision analysis and quantity estimates. It does not replace a licensed architect, structural engineer, quantity surveyor, or local municipal authority approval. All quantities must be field-verified before material procurement.

1. PROJECT & DRAWING METADATA

Project Name	Villa Residence G+1	Building Type	Residential
Floors	2	Unit System	METRIC
Target Area	180.0 m²	Soil Type	Not Provided
Drawing Scale	1:100	Concrete Grade	M20
Wall Thickness	230 mm	Overall Confidence	HIGH

2. CONSOLIDATED MATERIAL ESTIMATES (UNCERTAINTY-AWARE)

MATERIAL	ESTIMATED	RANGE	CONFIDENCE	CALCULATION BASIS
Cement (OPC/PPC 50kg Bags)	2,152.0 bags	1,980 - 2,367	MEDIUM	Concrete (557.4 bags) + Masonry (1019.8 bags)...
River Sand / M-Sand	336.8 m³	303 - 377	MEDIUM	Concrete (29.0m³) + Masonry (212.5m³) + Plast...
Coarse Aggregate (20mm & 10mm Graded)	341.0 m³	76 - 93	MEDIUM	RCC Concrete (58.1m³) + PCC Bed (26.8m³)
Modular Clay / Fly Ash Bricks	296,499.0 nos	272,779 - 326,149	HIGH	564.76 m³ net masonry x 500 bricks/m³ + 5% br...
Reinforcement Steel (Fe500 TMT Rebar)	5,472.0 kg	4,396 - 6,464	MEDIUM	Slabs (3391.0 kg at 0.8% vol) + Beams/Columns...
Vitrified Floor Tiles (600x600mm)	34.7 m²	33 - 38	HIGH	Net floor area 32.1m² + 8% cutting and breaka...
Emulsion Paint & Primer (2 Coats)	564.6 litres	508 - 649	MEDIUM	Total plaster & ceiling area 5645.9m² / cover...

3. DETAILED BILL OF QUANTITIES (CIVIL & FINISHES)

CODE	DESCRIPTION	QTY	UNIT	RANGE	CONF.
CIV-01	Earthwork excavation in foundation trenches	434.52	m³	304.2 - 630.0	LOW
CIV-02	Plain Cement Concrete (PCC 1:4:8) in foundation bed (100mm)	31.49	m³	28.3 - 36.2	MEDIUM
CIV-03	Reinforced Cement Concrete (M20) for Slabs, Beams, Columns	69.12	m³	63.6 - 77.4	MEDIUM
CIV-04	Brick Masonry in Cement Mortar (1:6) with Modular Bricks	564.76	m³	508.3 - 621.2	HIGH
CIV-05	Cement Plastering (12mm Internal 1:6 + 18mm External 1:4)	5285.90	m²	4915.9 - 5814.5	MEDIUM
CIV-06	Flooring with Vitrified Tiles including 8% wastage	34.67	m²	32.9 - 37.4	HIGH

4. DETECTED BLUEPRINT ANOMALIES & COMPLIANCE FINDINGS

ID	SEVERITY	FINDING & EVIDENCE	RECOMMENDED ACTION
ISSUE-001	MEDIUM	Potential Habitable Area Constraint: LIVING & DINING Measured area: 3.8 m² vs Recommended threshold: 9.5 m² <i>Ref: NBC 2016 Part 3 Clause 4.2.1 (Minimum Habitable Room Area)</i>	Review room layout with a licensed architect to confirm local municipal development control guidelines. Verification: Required
ISSUE-002	MEDIUM	Staircase Riser and Tread Detail Missing on 2D Plan No textual riser/tread callouts (e.g. '18 Risers @ 165mm') identified in staircase zone. <i>Ref: IBC Section 1011.5.2 & NBC 4.8.1</i>	Request sectional drawing / stair detailing schedule for precise concrete and reinforcement calculation. Verification: Required

5. LOCAL AI (GEMMA 2B) SYNTHESIS & REASONING

Executive Synthesis: Preliminary architectural evaluation performed for Residential project. Detected 2 potential dimensional or compliance considerations (0 require urgent review). Soil parameter was 'Not Provided', requiring geotechnical confirmation before finalizing foundation estimates.

Uncertainty Assessment: Overall analysis confidence is rated as HIGH. Confidence is constrained by 2D plan annotations and missing structural section schedules.

6. MISSING PARAMETERS & ENGINEERING ASSUMPTIONS

- Missing Required / Recommended Inputs:**
- **Soil Type:** Not Provided - *Foundation trench excavation, footing depth, and sub-base quantities cannot be verified without soil bearing capacity (SBC).*
- Applied Engineering Assumptions:**
- Excavation depth assumed at nominal 1.2m for medium dense soil; footing sizes subject to soil report.
 - Reinforcement steel estimated via standard IS 456 thumb rules (0.8% for slabs, 1.5% for framed elements).